1 A student investigates the current in a coil and a resistor connected in series, as shown in Fig. 1.1.

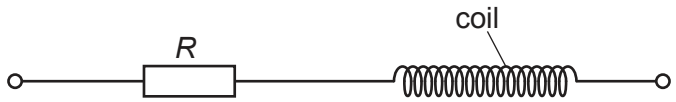

Fig. 1.1

The student connects a high-voltage d.c. power supply and a switch across the series combination.

When the switch is closed, it takes time t for the current in the resistor of resistance R to reach a maximum value. The time t is a few milliseconds.

There are a number of different unmarked resistors available.

It is suggested that the relationship between t and R is

$$t = \frac{KN^2A}{LR}$$

where N is the number of turns of wire on the coil, A is the cross-sectional area of the coil, L is the length of the coil and K is a constant.

Design a laboratory experiment to test the relationship between t and R.
Explain how your results could be used to determine a value for K.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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